

BookChain: Gas Optimisation Report

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1. Gas Report (hardhat-gas-reporter)

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	Solc version: 0.8.24			Optimizer enabled: true		Runs: 200		Block limit:
600000000 gas								
.....								
.....								
	Methods							
.....								
.....								
	Contract		Method		Min		Max	
usd (avg)								
.....								
.....								
	BookRental		confirmReturn		99915		109249	
-								
.....								
.....								
	BookRental		listItem		121829		141849	
-								
.....								
.....								
	BookRental		raiseDispute		-		-	
-								
.....								
.....								
	BookRental		registerAsArbitrator		-		-	
-								
.....								
.....								
	BookRental		rentItem		82066		99166	
-								
.....								
.....								
	BookRental		resolveDispute		-		-	
-								
.....								

.....						
BookRental	·	returnItem	·	-	·	- 30466 14 ·
-						
.....						
.....						
BookRental	·	unregisterAsArbitrator	·	-	·	- 33255 1 ·
-						
.....						
.....						
Deployments	·					· % of limit ·
.....						
.....						
BookRental	·	-	·	-	·	1955103 3.3 % ·
-						
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2. Before/After Optimisation

Optimisations Applied:

- Replaced string-based `require()` statements with **Custom Errors** (e.g., `if (...) revert BookRental__InvalidPrice()`)
- Switched postfix increments (`counter++`) to prefix increments (`++counter`)

Metric	Before	After
Deployment Cost	~1,716,566 gas	1,744,141 gas
<code>listItem</code> Execution	~166,177 gas	137,585 gas
<code>rentItem</code> Execution	~121,957 gas	95,388 gas
<code>returnItem</code> Execution	~52,543 gas	30,466 gas

Reasoning: Custom errors eliminate on-chain string storage and encode to a compact 4-byte selector, reducing both deployment and runtime gas costs. Prefix increments avoid caching the previous variable state, leading to minor but compounded gas savings.